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Experiment-25

Question

Write a C program that displays a menu and uses a **switch statement** to print the following **right-aligned patterns** selected by the user. Use **nested loops without arrays**. Each pattern should be implemented using a separate **parameterless function with no return value**.

1. Right-aligned increasing triangle
2. Right-aligned increasing number triangle
3. Right-aligned decreasing triangle
4. Exit

Algorithm

1. Start.
2. Declare `choice`.
3. Display the menu:
 - 1. Right-aligned increasing triangle
 - 2. Right-aligned increasing number triangle
 - 3. Right-aligned decreasing triangle
 - 4. Exit
4. Input the user's choice.
5. Use a `switch` statement.
6. If choice is 1, call `pattern1()`.
7. If choice is 2, call `pattern2()`.
8. If choice is 3, call `pattern3()`.
9. If choice is 4, exit the program.
10. Otherwise, display "Invalid choice".
11. Repeat the menu until the user selects Exit.
12. Stop.

Algorithm for Pattern 1

1. Input number of rows n .
2. For each row i from 1 to n :
 - Print the required spaces.
 - Print i asterisks.
3. Stop.

Algorithm for Pattern 2

1. Input number of rows n .
2. For each row i from 1 to n :
 - Print the required spaces.
 - Print natural numbers from 1 to i .
3. Stop.

Algorithm for Pattern 3

1. Input number of rows n .
2. For each row i from n down to 1:
 - Print the required spaces.
 - Print i asterisks.
3. Stop.

Test Case Table

C Program

```
#include <stdio.h>

void pattern1();
void pattern2();
void pattern3();

int main()
{
    int choice;

    do
    {
        printf("\n1. Right-aligned increasing triangle");
        printf("\n2. Right-aligned increasing number triangle");
        printf("\n3. Right-aligned decreasing triangle");
        printf("\n4. Exit");

        printf("\nEnter your choice: ");
        scanf("%d", &choice);
```

```

switch(choice)
{
    case 1:
        pattern1();
        break;

    case 2:
        pattern2();
        break;

    case 3:
        pattern3();
        break;

    case 4:
        printf("Exiting program...\n");
        break;

    default:
        printf("Invalid choice\n");
}

} while(choice != 4);

return 0;
}

void pattern1()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

void pattern2()
{
    int n, i, j;

```

```

printf("Enter number of rows: ");
scanf("%d", &n);

for(i = 1; i <= n; i++)
{
    for(j = 1; j <= n - i; j++)
        printf(" ");

    for(j = 1; j <= i; j++)
        printf("%d ", j);

    printf("\n");
}
}

void pattern3()
{
    int n, i, j;

    printf("Enter number of rows: ");
    scanf("%d", &n);

    for(i = n; i >= 1; i--)
    {
        for(j = 1; j <= n - i; j++)
            printf(" ");

        for(j = 1; j <= i; j++)
            printf("* ");

        printf("\n");
    }
}

```

Compilation And Execution

```
(kali㉿kali)-[~]  
$ vi hdoc26.c  
  
(kali㉿kali)-[~]  
$ gcc hdoc26.c -o hdoc26  
  
(kali㉿kali)-[~]  
$ ./hdoc26  
  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 1  
Enter number of rows: 4  
  *  
 * *  
* * *  
* * * *  
  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 2  
Enter number of rows: 4
```

```
* * * *  
 * * *  
  * *  
   *  
  
1. Right-aligned increasing triangle  
2. Right-aligned increasing number triangle  
3. Right-aligned decreasing triangle  
4. Exit  
Enter your choice: 4  
Exiting program...
```